

AMARJEET VISHWAKARMA

Low-Latency C++ Software Developer | Trading Systems

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

C++ software developer with 2+ years building ultra-low-latency trading infrastructure in production. Specializes in lock-free concurrency, zero-copy networking, and FIX protocol engines processing 3M+ trades daily across NSE, BSE, MCX, IIBX, and crypto exchanges with sub-100-microsecond order-to-exchange latency.

SKILLS

Languages: C++ (C++11/14/17/20)

Systems & Low-Latency: Multithreading, Lock-free Programming, Custom SPSC/MPMC Queue Design, Memory Management, Cache-friendly Design, Zero-copy Techniques, Custom Allocators, Core-pinned Network Publishers

Networking: TCP/IP, UDP, Multicast, POSIX Sockets, Winsock, FIX Protocol (4.2/4.4)

Concurrency: Thread Pools, Atomic Operations, Mutex-free Design, SPSC/MPMC Queues

Databases & Messaging: MySQL, RabbitMQ

Tools & Platforms: Linux (Ubuntu, RHEL), Windows, GDB, Valgrind, perf, Git, CMake

Design & Architecture: Low Level Design (LLD), Trading Systems, OMS Architecture, Order Lifecycle Management

EXPERIENCE

Software Developer - Finsol Technologies,

Dec 2023 - Present | Noida, UP

Multi-Exchange Trading Gateway & Order Execution

- Architected** a high-performance multi-exchange trading gateway in C++, processing 3M+ trades daily at 99.9% uptime across NSE, BSE, MCX, IIBX, and crypto (ByBit).
- Designed** an ultra-low-latency order execution system achieving sub-100-microsecond order-to-exchange latency using C++, Boost.Asio, and lock-free queues.
- Built** market data handling for both snapshot and tick-by-tick (TBT) feeds across multiple exchanges, reducing dissemination latency by 35% through zero-copy techniques and efficient memory pooling.

FIX Engine, DropCopy & Messaging Infrastructure

- Implemented** a FIX 4.2/4.4 engine with custom parsers, achieving sub-50-microsecond message processing latency.
- Developed** a real-time DropCopy module for trade reconciliation and regulatory audit compliance.
- Integrated** RabbitMQ with message persistence and failover for reliable component communication.
- Integrated** AES-128-GCM and AES-256-GCM encryption via OpenSSL with minimal latency overhead.

Performance Engineering & Reliability

- Improved** inter-thread performance by cutting lock contention 60% using Boost.Lockfree SPSC/MPMC queues.
- Optimized** latency-critical hot paths using GDB, Valgrind, and perf, reducing CPU cycles by 40%.
- Minimized** heap allocations in critical paths by 80% through custom allocators and memory pooling.
- Built** a high-resolution logging framework with microsecond-level timestamps for monitoring and post-trade analysis; strengthened system stability with automated connection recovery.

Software Developer Intern - Finsol Technologies,

Jun 2023 - Dec 2023 | Noida, UP

- Engineered** cross-platform POSIX/Winsock socket modules with heartbeat-based reliability.
- Delivered** non-blocking, low-latency logging for real-time financial transaction monitoring.
- Worked** on Order Management System (OMS) architecture and full order lifecycle handling across exchange protocols.

PERSONAL PROJECTS

FIX Protocol Parser & Visualizer - C++, React, Docker

- Full-stack FIX 4.2/4.4 parser: C++/Crow backend with 200+ tag reference, validation, and repeating-group detection.
- React frontend with message visualization, order-lifecycle correlator (ClOrdID/OrigClOrdID), and interactive tag lookup.
- Deployed independently on Render via Docker with live interface and shareable links.

Live demo: fix-parser-frontend.onrender.com | Source: [fix-parser-backend](#) · [fix-parser-frontend](#)

PCAP Multicast Replay Console - C++, React, PCAP, HTTP

- C++ HTTP backend (cpp-httplib) with PCAP parsing, packet filtering, and multicast UDP routing for channel-based replay.
- React dashboard with packet statistics (Chart.js), real-time replay controls (start/pause/stop/speed), and interactive packet preview.
- Loop-based replay with configurable limits, channel management UI, and backend heartbeat detection.

Live: <https://amar2796.github.io/pcapreplay/> | GitHub: <https://github.com/amar2796/pcapreplay>

EDUCATION

PG-DAC - PG Diploma in Advanced Computing, CDAC Noida,

Mar 2023

B.Tech, Information Technology - CCS University, Meerut,

2015 - 2018

Cumulative Percentage: 73% | **Additional:** Solved 250+ DSA problems on LeetCode and HackerRank.